Independent audit of the continuum two-body package: conditional rectangle agreement, phase mismatch and V1

Quantum Boussinesq solitons project

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Abstract

(c) The leading rectangle is indeed $-(15\sqrt{2}-20)/\pi = -0.386174647502482\ldots$. Its construction, however, imports the Toda endpoint-product identity: it tests a one-body coefficient inside a specified Toda frame, rather than deriving that frame from the continuum determinant. The printed subleading rectangle, profile sign, stability band, one classical series coefficient, general tadpole expansions and numerical tables contain errors, despite all three package scripts printing their success marker. Independent quadrature, inverse Laplace transforms and unconstrained refits support the conditional Toda numbers. The continuum canonical candidate differs from the Toda directional antisymmetric coefficient by a nonconstant function of the ratio; the equality of their rational terms does not repair this discrepancy. **(d)** V1 is not a completed validation. The physical comparison remains conditional on a common quantum endpoint and momentum dictionary, and on a defined bad-equation determinant. No scheme for the PI's good equation is selected by this audit.

1 Scope, methods and verdict table

Labels retain their repository meanings: (a) literature proof, (b) literature heuristic, (c) project derivation or numerical evidence, and (d) unestablished interpretation. "Confirmed" below concerns the precise algebra or numerical test named, not an exact physical S-matrix. Numerical evidence is not a uniform limit theorem. The verdict table covers all twelve rows of the package's claim summary, with additional rows for conventions, its untested assertions, the proposed rational-term identification and V1.

(c) Read inputs include `AGENTS.md`, `README.md`, `twobody_cont.tex` and its PDF, all three package scripts and saved JSON, the format and relevant conclusions of `audit_phase2i.tex`, `twobody_toda.tex` and its grids, `audit_frame_dictionary.tex`, `audit_onebody_scheme.tex`, the relevant determinant/measure audits, `oneloop_phase.tex`, `onebody_dispersion_test.tex`, `errata.tex`, and the re-ranked next steps in `report.tex`. Only internal sources are cited; no external full text or abstract was newly read, and no inherited arXiv/DOI is represented as newly verified.

(c) All new checks are in `scripts/audit_twobody_cont.py`. The checker imports no package or earlier audit module. It uses the unfactored Hirota polynomial, profile differentiation, an integral representation of the continued energy, direct phonon quadrature, and independent symbolic division. Algebraic/integral checks use 70-digit arithmetic, with 110 digits for small-amplitude cancellation. The independent Toda calculation uses de Hoog inversion and nested Clenshaw–Curtis integration, replacing the package's Talbot/Gauss calculation. It is a fresh solution of the stated half-line model, not a new finite-sea simulation. Saved finite-sea data are independently refitted, with their original float64 precision retained. Refutations are encoded as assertions that a discrepancy exists. The input hashes are checked at the end. No existing repository file or git state was changed; all new paths begin with `audit_twobody_cont`.

(c) A concurrently supplied `audit_twobody_toda.tex` was also inspected. Its verdict confirming the displayed directional closed form and qualifying its physical interpretation is consistent

with the present independent checks. Its numerical table, however, prints 2.1670489477 at $r = 0.5$ for a formula whose direct value is 1.530527813279; that decimal is not used here. Its wording identifying this hyperbolic validation with the good equation is not adopted: the fixed project good/bad convention takes precedence. The corrected four-term reference source is retained; the older frame-audit source error is not an input to any calculation here.

ID	Claim	Verdict	Evidence / limitation
0	Printed good/bad and Toda conventions	Refuted (c)	Profile, L_0, sound speed and stable-band signs fail direct checks.
1	Exact bad Hirota factor, positivity, action gradients	Confirmed (c)	Unfactored polynomial, positive hyperbolic factors, four signed sectors.
2	Classical small-amplitude expansions	Qualified (c)	Action and head-on terms survive; printed overtaking ε^4 coefficient fails.
3	Bad determinant integrals and claimed $c_{2,\mathrm{det}}$ verification	Refuted (c)	No such integral or nine-root test is in the scripts; real-frequency whole-line claim is false. Formal continuation not specified.
4	Matched energy, canonical slope and Jacobian	Qualified (c)	Conditional trace algebra survives; bare momentum β^5 misprint; Jacobians are different coordinates.
5	Canonical and mode frame-shift formulas	Confirmed (c), conditional	Direct differentiation and small-amplitude limits; physical momentum selection remains (d).
6	Sound scaling, virial identity and small shift	Confirmed (c)	Counterterm variation only; interpreting it as a complete third prescription is qualified (d).
7	Kinetic norm and finite constant	Confirmed (c)	Direct finite-box integral and independent Fourier moment; asymptotic, not exact at finite R.
8	Projector tails and claimed commutator implication	Qualified (c)/(d)	Boundary issue survives; Poisson brackets alone do not prove that commutator.
9	Tadpole sensitivity and higher local counterterms	Qualified (c)	Leading power counting survives; printed general series omit tadpole terms.
10	KdV inversion and exchange antisymmetry	Confirmed (c), conditional	Direct arctangent quadrature and exchanged signed classical shifts. Not raw-phase antisymmetry.
11	Rectangle prediction and Toda agreement	Qualified (c)	Leading limit confirmed; printed $1/Y$ term refuted; endpoint structure borrowed from Toda.
12	Unsettled determinant, constants and Wronskian selection	Qualified (d)	Functional problem open; finite-dimensional Wronskian already factorises in earlier audit.
13	Grids, printed tables and claimed precision	Refuted (c) as stated	Saved algebraic components reproduce; tables disagree, most JSON values are float64, determinant components absent.

ID	Claim	Verdict	Evidence / limitation
14	All scripts pass and verify every (c) claim	Qualified (c)	All pass; claimed coverage is false.
15	Rational Darboux term identifies Toda dressing	Qualified (c)/(d)	Algebraic equality confirmed; same physical object not established; remainder is not I or J.
16	V1 completed continuum–Toda validation	Refuted as completion (c)/(d)	Explicit grid discrepancies; physical equivalence remains inconclusive.

2 Conventions and classical checks (0–2)

The positive profile is required (c; `audit_twobody_cont.py`, `symbolic`). Twice integrating the decaying travelling-wave equation gives

$$k^2u - u_{xx} - 3u^2 = 0, \qquad v^2 = 1 + k^2.$$

The positive $u_+ = k^2\,\mathrm{sech}^2(k\xi/2)/2$ solves it exactly. The printed negative profile gives residual $-6u_+^2$, and cannot equal $2(\log(1+e^{k\xi}))_{xx}$. The consistent potential has $\phi = C + k\tanh(k\xi/2)$; its $\Pi = \int \phi_t = -2vk$ agrees with the printed charge. The printed negative potential would instead reverse this charge. Furthermore

$$vP_0 - E_0 = +\frac{2}{15}\sinh^5\beta, \qquad \frac{d(vP_0 - E_0)}{dv} = P_0.$$

The note prints the opposite sign, although its phase script checks the positive sign. Squaring the potential tangent conceals the profile-sign error in the kinetic norm.

(c) Linearisation of the bad PDE yields $\omega^2 = q^2(1-q^2)$. The stable vacuum band is $0 < |q| < 1$, not $|q| > 1$; the instability is at short wavelengths. At infinity a localized soliton has this same laboratory dispersion. Its velocity does not replace the vacuum band edge by $\sqrt{1+\sinh^2\beta}$. Direct continuation of the Hirota coefficient at $(\beta, q) = (0.7, 1.3)$ has modulus different from one (approximately 1.839), so there is no real whole-line scattering phase of the asserted type. Keeping $t = \sqrt{1+q^2}$ is not the bad real-frequency dispersion. These errors contradict the controlling one-body scheme audit.

(c) With $b = e^{-a/2}$ the Toda elastic constant is b^2, hence the physical sound speed is ab, not the printed $a\sqrt{b}$. Differentiating the stated lattice charges gives $dE_T/dP_T = ab\sinh\kappa/\kappa \to ab$. The action/Planck map $h = 3b\hbar_B$ and $y = \beta^2/\hbar_B$ consistently identify dimensionless phases at leading continuum order. They are not an exact finite-amplitude equivalence: substituting $\kappa = \sqrt{3}\beta$ in the lattice momentum yields

$$\frac{aP_T}{6\sqrt{3}b} - P_{0,B} = -\frac{7}{15}\beta^5 + O(\beta^7).$$

Thus "identically" must be restricted to the continuum action normalization.

(c) No defect scattering or spectral calculation enters this package's actual formulas. The errata dictionary remains $s = \sigma_{\mathrm{vac}}$ for scattering and $\eta\sigma_{\mathrm{vac}}$ for the spectrum. The CZ field/action dictionary is inherited as $u_{\mathrm{CZ}} = -3\phi$, $L_{\mathrm{CZ}} = 9L$; it is not used to choose a new convention. There is no defect-s test here to certify.

Exact classical core confirmed (c; `continuum_checks`). Using $P_-(K, W) = W^2 - K^2 - K^4$ directly, rather than the package's reduced implementation, reproduces its factored coefficient

and all four signed action-gradient identities. Positivity follows explicitly by putting $p = \beta_1 + \beta_2$, $m = \beta_1 - \beta_2$:

$$A_{\rm ov} = \frac{\sinh^2(m/2)}{\sinh^2(p/2)} \frac{1 + 2\cosh p}{1 + 2\cosh m},$$
$$A_{\rm ho} = \frac{\cosh^2(p/2)}{\cosh^2(m/2)} \frac{2\cosh m - 1}{2\cosh p - 1}.$$

Both are positive off the equal-amplitude overtaking diagonal. The oriented algebraic phase uses the package's label convention; calling its extension to every ordering a physical collision phase needs that orientation convention to be retained.

(c) Formal division of the unfactored polynomial gives the correction

$$\frac{A_{\rm ov}(\varepsilon, r\varepsilon)}{A_0(r)} = 1 + r\varepsilon^2 + \left[\frac{r^2}{2} - \frac{13r(1+r^2)}{60}\right]\varepsilon^4 + O(\varepsilon^6). \tag{1}$$

At $r = 1/2$ the coefficient is $-1/96$, rather than $+1/8$. The script only checks the ε^2 term. The action expansions through ε^4, head-on coefficient through ε^4, and canonical H, D expansions through ε^2 are independently confirmed; the error in (1) first affects the overtaking action at the next order.

3 Phonon integrals and one-body inputs (3–4)

Verified function versus uncomputed determinant (c). The scale-free function is independently obtained from a direct phase integral, not from evaluation of the dilogarithm against itself:

$$I(a) = -\frac{4}{\pi}\int_0^{\pi/2} \arctan\Big(\frac{a}{\tan t}\Big)\, dt = \frac{4}{\pi}\int_0^a \frac{\log x}{1 - x^2}\, dx. \tag{2}$$

Its dilogarithmic expression and $I(a) + I(1/a) = -\pi$ follow by differentiating and fixing the $a \downarrow 0$ constant; both are independently checked. The symmetric endpoint offset is retained: $J = I + \pi/2$ in the ++ chamber. The old refuted claim of raw-phase antisymmetry is not used.

(c) There is also an ordering issue. In the real good-contour integral $I_{12} = \int \theta_1 \theta_2' / (4\pi)$, the small positive-$q$ phases are $\theta_j = 4\arctan(k_j/q)$ and the negative-q leading phases vanish. Equation (2) therefore has $a = k_1/k_2$, whereas this package sets $r = \beta_2/\beta_1$. A direct quadrature of the full good phonon expression at $(\alpha_1, \alpha_2) = (10^{-5}, 5\cdot 10^{-6})$ gives -2.000671508529, approaching $I(2) = -2.000671508540$, not $I(1/2) = -1.140921145050$. The continued small-root ratio also remains $\sinh\beta_1/\sinh\beta_2$. A reversal or redefinition of the directional integral must therefore be specified before identifying the package's $I(r)$ with that source. This test does not select a contour for the unstable bad theory. The grid comparisons below grant the package its own $I(r)$ definition, so this additional issue cannot manufacture the reported mismatch.

(c) Inspection of all three complete scripts finds no computation of the bad finite-amplitude I_{12}, no two-dimensional phonon quadrature, and no nine-term dilogarithm test of the claimed $c_{2,\rm det}^{\rm ov}$ or $c_{2,\rm det}^{\rm ho} = -8r/\pi$. The phase script verifies $I(r)$ against a one-dimensional integral for χ_2 and the classical/frame expansions. Its output even records a different, logarithmic head-on expression, without evaluating it. The grid script inserts $-8r/\pi$ as data. Thus the asserted 60-digit verification and the associated error budget are refuted; a particular continued finite-amplitude determinant coefficient remains unestablished (d). An $O(\hbar_B)$ counterterm does not change the leading quadratic operator in loop counting (c), but this observation supplies neither its boundary domain nor the missing integral.

One-body algebra survives conditionally (c; `symbolic`, `continuum_checks`). Integrating $dF_{\rm ac}/dk = 2\sqrt{3}k\arctan(2k/\sqrt{3})/\pi$ from $k = 0$ and differentiating with respect to velocity independently reproduces $F_{\rm matched}$, $C = 1/(2\pi)$, $p_{\rm can,matched}$, and the displayed canonical momentum

Jacobian. In the bare momentum series the correct β^5 coefficient is $-8\epsilon/(45\pi)$, not the printed $-4\epsilon/(15\pi)$; the code uses the correct value. These are properties of the continued trace, not a construction of a physical bad-equation determinant. No Toda two-body number sets the energy's integration constant: $F_{\rm ac}(0) = 0$ and the vacuum-fixed counterterm are the inputs.

(c) Direct quadrature of the actual stable-band transmission, retaining the specified branch and its normal-ordering term, confirms

$$F_{\rm band} = F_{\rm ac} - \frac{\sqrt{3}}{2}(\sinh^2\beta + 3/4).$$

Thus its threshold constant is correct, but the package's explanation as integration over $|q| > \sqrt{1+\sinh^2\beta}$ is false. The checks use the real band $q = \sin t$, $|t| < \pi/2$. No new physical regularisation is selected.

4 Frame choices, local terms and the projected measure (5–10,12)

Three expressions, not three proved equivalent implementations (c)/(d). Direct charge differentiation verifies $\gamma = (d\Pi/d\beta)/(dP_0/d\beta)$ and $p_{\rm can} = -\epsilon F'/\sinh\beta$. Hence $F = 2C\beta^3/3 + O(\beta^5)$ gives $p_{\rm can} = -2\epsilon C\beta + O(\beta^3)$ and

$$H_{\rm ov} \to 0, \quad D_{\rm ov} \to -4C(r^{-1} - r), \qquad H_{\rm ho} \to 0, \quad D_{\rm ho} \to 4C(r^{-1} + r).$$

The stated mode coefficients are checked by continuing the elementary one-body root integral $\sum_j c_j \sin\phi_j \,\mathrm{atanh}(\cos\phi_j)/(2\pi)$, explicitly as a formal continuation, and reproduce its logarithmic slope and constant. Substitution in the geometric shifts verifies its nonzero $H_{\rm ov}$ and logarithmically divergent D.

(c) Direct rescaling of the saddle gives $(S, P, E, v) \mapsto (c_s^3 S, c_s^4 P, c_s^5 E, c_s v)$ and $5E_0 - 4vP_0 = 2\sinh^3\beta/3$. Differentiating $c_s^3\delta_0(P_1/c_s^4, P_2/c_s^4)$ gives the printed sound counterterm insertion, which vanishes as ε^2. This is the *change due to the counterterm*; it does not compute or erase the pre-existing bare one-loop Darboux shift. Consequently the inference "sound insertion tends to zero, therefore the full answer is phonon-only" does not follow. A proposed Wronskian cancellation would be another input (d). Likewise, $e - vp = F$ is invariant under $(e, p) \mapsto (e + v\delta p, p + \delta p)$; it cannot choose the momentum entering a two-body phase. **(d)** Canonical momentum is natural for a completed collective action, but that action's boundary-compatible reduction is not supplied here and is not forced by the vacuum match.

Kinetic norm confirmed; projector inference qualified (c)/(d). For the printed tangent's square, direct finite-box integration and $-\partial_q^2[\frac{4}{3}\Gamma(2 + iq/2)\Gamma(2 - iq/2)]_{q=0}$ independently give

$$\int_{\mathbb{R}} z^2 \,\mathrm{sech}^4 z \, dz = \frac{\pi^2 - 6}{9}.$$

This verifies the positive finite part $v^2(\pi^2 - 6)\dot{v}^2/(9k^3)$ alongside the linear divergence. The claimed "exact finite-R" formula actually has an explicit exponential remainder and is asymptotic. The finite pairing dP_0/dv concerns a velocity tangent, not a unit canonical-P tangent; tangent normalization matters. For coordinate tangents and the stipulated $\Omega = \int d\pi \wedge d\phi$, direct integration gives $\Omega(\partial_X, \partial_v) = -dP_0/dv$; reversing their order gives the positive sign. The package's $\Omega(\xi_X, \xi_P) = +dP_0/dv$ therefore also requires explicit tangent normalization and orientation, which its scalar derivative check does not test.

(c) The existence of potential tails and the need to control the operator domain are consistent with the earlier audits. The assertion that nonzero canonical Poisson brackets imply $[\partial_t - v\partial_\xi, \mathcal{P}] \neq 0$ is not a valid implication. A constant symplectic projector can commute with

a block-preserving evolution while its projected canonical pair has nonzero bracket; the checker supplies this elementary matrix counterexample. No explicit bad field projector or Schur determinant is constructed by the package's script, which merely writes their descriptions into JSON. **(d)** The actual functional boundary problem remains open.

(c) The package also overlooks a controlling result of `audit_projected_measure.tex`. Independently recomputing

$$W = \begin{pmatrix} C & D \\ -D & 0 \end{pmatrix}, \qquad D = \mathrm{diag}(d_1, d_2), \quad C = \begin{pmatrix} 0 & c \\ -c & 0 \end{pmatrix}$$

gives $\det W = (d_1 d_2)^2$ for every tail pairing c. There is no tail-dependent finite-dimensional symplectic Jacobian available to cancel D_{12}. An additional functional determinant is still possible (d), but it cannot be inferred from that already-factorised Wronskian.

Counterterm statement: correct power counting, incorrect general series (c). Writing $t = c_{\mathrm{tad}}$, the actual series are

$$\frac{F}{|P_0|} = \frac{3t}{\beta^2} + \frac{2}{\pi} + \lambda - \frac{5}{2}t + \left(\frac{53t}{40} + \frac{3}{2}c_5 - \frac{\lambda}{2} - \frac{23}{15\pi}\right)\beta^2 + O(\beta^4), \tag{3}$$

$$\frac{p_{\mathrm{can}}}{\epsilon} = -\frac{2t}{\beta} + \left(-\frac{2t}{3} - \frac{4}{\pi} - 2\lambda\right)\beta + \left(\frac{2t}{45} - 5c_5 - \frac{4\lambda}{3} - \frac{8}{9\pi}\right)\beta^3 + O(\beta^5). \tag{4}$$

The printed equations omit the regular tadpole terms and partially specialize λ without stating the corresponding restriction in both equations. The scripts check selected coefficients, setting $t = 0$ before testing the regular ratio. They never assert the displayed general formulas.

(c) With $t = 0$ and an analytic *one-body restriction* $\Delta F = c_5 \sinh^5 \beta + O(\beta^7)$, $\Delta(F/|P_0|) = O(\beta^2)$ and $\Delta p = -5\epsilon c_5 \beta^3 + O(\beta^5)$. Since both geometric shifts are at most $O(\varepsilon^{-1})$ at fixed $0 < r < 1$, their induced H, D corrections are $O(\varepsilon^2)$. This confirms exactly the requested invariance of C and these $O(1)$ frame shifts. It does not prove absence of every independent two-body local insertion, nor uniformity as $r \to 0, 1$, nor exclusion of nonanalytic or nonlocal boundary terms. The tadpole cancels between separated charge sectors by mass additivity; its vacuum coefficient is an inherited condition, not independently fixed by this package.

(c) Exchange antisymmetry of H, D and the sound insertion follows by swapping both amplitudes and orientation signs; it is also checked directly at finite amplitude. The raw phonon integral has the nonzero symmetric endpoint offset. All such algebraic antisymmetries retain the definition of the oriented phase.

5 The rectangle: what agrees, what was imported (11)

Independent four-corner expansion (c; `rectangle`). Let $f(y) = \sqrt{y}/u'(y)$. Conditional on the *Toda* endpoint-product reduction, the rectangle is

$$\mathcal{D}_0(Y) = 8[f(2Y) - f(Y)][f(Y/2) - f(Y/4)] - 8[\sqrt{2Y} - \sqrt{Y}][\sqrt{Y/2} - \sqrt{Y/4}]. \tag{5}$$

Additive one-body terms cancel. Independently expanding all four corners, with $f(y) = \sqrt{y} + C/\sqrt{y} + B/y^{3/2} + \cdots$, gives

$$\mathcal{D}_0(Y) = -2C(15\sqrt{2} - 20) + \frac{8C^2(\sqrt{2} - 1)^2 + (102 - 85\sqrt{2})B}{Y} + \cdots. \tag{6}$$

The leading limit is confirmed, including the earlier frame audit's exact number and the bare value four times as large. The printed subleading claim is false: even $u' = 1 - C/y$ generates $B = C^2$ on inversion. The stated weaker remainder $O(y^{-3/2})$ in u' also permits an $O(Y^{-1/2})$ rectangle term, larger than the asserted $1/Y$ correction. For the corrected Toda expansion $u' =$

$1-C/y-d_0/y^2+\cdots$, $d_0 = 1/48-C^2$, so $B = 1/48$ and the $1/Y$ coefficient is $-0.344568583099\ldots$, not the package's positive $8C^2(\sqrt{2}-1)^2$. Its script inserts $f = \sqrt{y} + C/\sqrt{y}$ and treats that truncation as if it determined the next order.

(c) The leading value differs from the quoted finite-Y estimate $-0.395751991872 \pm 0.021668942196$ at $Y = 32$ by 0.009577344370. That arithmetic agreement is genuine; the uncertainty is an edge-fit spread, not a confidence interval and not a bound on the finite-Y bias. The independent inverse-transform rectangle at finite Y is tabulated below.

Circularity and identification (c)/(d). The energy coefficient was not fitted to a Toda two-body target. Its sole lattice matching input is vacuum elasticity. However, the rectangle formula itself, the endpoint identity and the finite-Y Jacobians used in the package comparison come from Toda; the code explicitly reads `frame_analysis.json`. The package supplies no continuum two-soliton determinant or endpoint operator calculation yielding (5). Consequently the result is a useful conditional one-body consistency check inside a borrowed two-body frame, not an independently established common continuum–Toda dictionary.

(c) In Toda the inverse-rapidity Jacobian coefficient is $A = 2(s_1 - C)$; it happens to equal $C = 1/(2\pi)$ in the matched units. A common sound-line shift changes s_1 and C together and leaves A unchanged. The continuum quantity $dP_{\rm phys}/dP_0$ is not, just from its name, du/dy for the Bethe rapidity. **(d)** Equating them requires the missing operator/coordinate map. Agreement of their first coefficients is evidence for such a map, not its construction.

6 Testing the rational-term lead and V1 (15–16)

A shared function is not yet a shared observable (c). Grant the continuum package its ordered r convention. Set

$$L_r = \frac{2}{\pi}\log r \log\frac{1-r}{1+r}, \qquad D(r) = \frac{2}{\pi}(r - r^{-1}).$$

Its canonical candidate and the conditional Toda result obey

$$C_{\rm ov}(r) = J(r) + D(r) = \frac{\pi}{2} - \frac{4}{\pi}\chi_2(r) - L_r + D(r), \tag{7}$$

$$\vartheta_N(r) = \pi - \frac{4}{\pi}\chi_2(r) + D(r), \tag{8}$$

$$C_{\rm ov}(r) - \vartheta_N(r) = -\frac{\pi}{2} - L_r. \tag{9}$$

Independent arctangent and regular-convolution integrals establish these relations without feeding either target into a fit. In particular the rational terms are identically equal, including sign, on the stated ordered chamber. But D is the continuum fixed-parameter to fixed-canonical-momentum conversion; Toda's term arises when its *directional* residual is antisymmetrised with its endpoint/Jacobian product. They have not been shown to be the same physical mechanism or frame transformation.

(c) The proposed remainder $\pi - 4\chi_2(r)/\pi$ is not the continuum raw integral I or antisymmetric integral J. It equals $I + H_{\rm mode} + \pi$, or $J + H_{\rm mode} + \pi/2$, as functions on this chamber. Combining that mode-only Hirota correction with $D_{\rm can}$ would splice two different momentum prescriptions: the same $p_{\rm mode}$ also generates a divergent Darboux term. A constant Maslov shift alone cannot remove L_r. **(d)** Thus the chi-function resemblance is a useful lead, but it does not identify the remainder as the physical dressing contribution still to be supplied by the continuum.

(c) Reference terms are substantial. The Toda grid defines $R_{12} = [\Phi(p_1, p_2) - \Phi(p_1, e_2) - \delta_T - C_{12}]/h$ and $\vartheta_N = (R_{12} - R_{21})/2$. It does not define a Gamma-relative scalar phase. Its endpoint-antisymmetric representative instead contains, relative to the classical phase, $2\sqrt{\pi Y}(1-r)$ in overtaking and $-2\sqrt{\pi Y}(1+r) + \pi + D_{\rm ho}$ in head-on, before lower corrections. Those different

references cannot be silently discarded. In the directional head-on comparison Toda tends to zero, while the continuum canonical candidate tends to $D_{\mathrm{ho}} = 2(r + r^{-1})/\pi$.

Grid comparison and uncertainties (c). The following tables list every common finite-sea point (15 overtaking and 15 head-on), plus all 17 common half-line points and the common ratio limits. The continuum column is its claimed $O(1)$ canonical function. At finite Y the Toda column includes inverse-Y corrections; these are shown as bias toward its separately tested limit, not reclassified as numerical error. Differences are diagnostic differences between the stated representatives. They are *not* measurements of a physical theory discrepancy until a common dictionary exists.

Table 2: (c) Common ratio limits. Toda directional head-on limit is zero.

r	D_{ov}	C_{ov}	$\vartheta_N(\infty)$	$C_{\mathrm{ov}} - \vartheta_N$	C_{ho}
0.25	-2.387324146	-1.587924848	0.433696751	-2.021621599	2.705634033
0.40	-1.336901522	-0.779277114	1.285772454	-2.065049569	1.846197340
0.50	-0.954929659	-0.525054477	1.530527813	-2.055582290	1.591549431
0.60	-0.679061091	-0.358489335	1.663132265	-2.021621599	1.443004817
0.75	-0.371361534	-0.189054656	1.738123625	-1.927178281	1.326291192
0.90	-0.134397507	-0.067364254	1.700929103	-1.768293357	1.280313098

Table 3: (c) Rectangle comparison; the last column uses Equation (6) with $B = 1/48$.

Y	Saved Toda	Edge spread	Fresh half-line	Leading limit	Corrected $1/Y$
4	-0.504362278	3.65×10^{-4}	-0.504364735	-0.386174648	-0.472316793
8	-0.434982726	1.32×10^{-3}	-0.435003014	-0.386174648	-0.429245720
16	-0.408705387	5.20×10^{-3}	-0.408875684	-0.386174648	-0.407710184
32	-0.395751992	2.17×10^{-2}	-0.397213133	-0.386174648	-0.396942416

Table 4: (c) All 30 common finite-sea points. Difference is continuum minus Toda. Error columns retain the package's separate diagnostics, not physical error bounds.

Y	r	Ch.	Continuum	Toda anti	Difference	Δ_h^{pkg}	$\Delta_{\mathrm{mesh}}^{\mathrm{pkg}}$
4	0.40	ov	-0.779277	0.760177	-1.539454	1.81×10^{-9}	1.04×10^{-8}
4	0.40	ho	1.846197	-0.000000	1.846197	5.75×10^{-8}	7.51×10^{-9}
4	0.50	ov	-0.525054	1.236044	-1.761099	1.78×10^{-8}	1.03×10^{-8}
4	0.50	ho	1.591549	-0.000000	1.591549	9.40×10^{-8}	7.52×10^{-9}
4	0.60	ov	-0.358489	1.474666	-1.833156	2.77×10^{-8}	1.02×10^{-8}
4	0.60	ho	1.443005	-0.000000	1.443005	1.35×10^{-7}	7.51×10^{-9}
8	0.40	ov	-0.779277	1.045501	-1.824778	2.13×10^{-8}	1.27×10^{-8}
8	0.40	ho	1.846197	-0.000000	1.846197	3.57×10^{-7}	7.61×10^{-9}
8	0.50	ov	-0.525054	1.402228	-1.927282	1.32×10^{-7}	1.27×10^{-8}
8	0.50	ho	1.591549	-0.000000	1.591549	6.48×10^{-7}	7.59×10^{-9}
8	0.60	ov	-0.358489	1.576517	-1.935006	2.21×10^{-7}	1.26×10^{-8}
8	0.60	ho	1.443005	-0.000000	1.443005	9.77×10^{-7}	7.69×10^{-9}
16	0.40	ov	-0.779277	1.178451	-1.957728	2.73×10^{-7}	1.64×10^{-8}
16	0.40	ho	1.846197	-0.000000	1.846197	2.37×10^{-6}	7.95×10^{-9}
16	0.50	ov	-0.525054	1.469720	-1.994774	9.83×10^{-7}	1.70×10^{-8}
16	0.50	ho	1.591549	-0.000000	1.591549	4.64×10^{-6}	7.96×10^{-9}
16	0.60	ov	-0.358489	1.621161	-1.979650	1.71×10^{-6}	1.74×10^{-8}

Table 4: (c) All 30 common finite-sea points. Difference is continuum minus Toda. Error columns retain the package's separate diagnostics, not physical error bounds.

Y	r	Ch.	Continuum	Toda anti	Difference	Δ_h^{pkg}	$\Delta_{\mathrm{mesh}}^{\mathrm{pkg}}$
16	0.60	ho	1.443005	-0.000000	1.443005	7.11×10^{-6}	8.09×10^{-9}
32	0.40	ov	-0.779277	1.234183	-2.013460	3.53×10^{-6}	2.38×10^{-8}
32	0.40	ho	1.846197	-0.000001	1.846198	2.47×10^{-5}	8.92×10^{-9}
32	0.50	ov	-0.525054	1.500789	-2.025844	7.24×10^{-6}	2.57×10^{-8}
32	0.50	ho	1.591549	-0.000001	1.591551	4.21×10^{-5}	9.20×10^{-9}
32	0.60	ov	-0.358489	1.642444	-2.000933	1.30×10^{-5}	2.67×10^{-8}
32	0.60	ho	1.443005	-0.000003	1.443007	6.93×10^{-5}	9.55×10^{-9}
64	0.40	ov	-0.779277	1.260419	-2.039696	4.27×10^{-5}	3.84×10^{-8}
64	0.40	ho	1.846197	-0.000014	1.846211	2.51×10^{-4}	1.27×10^{-8}
64	0.50	ov	-0.525054	1.515810	-2.040864	7.09×10^{-5}	4.25×10^{-8}
64	0.50	ho	1.591549	-0.000026	1.591576	4.09×10^{-4}	1.40×10^{-8}
64	0.60	ov	-0.358489	1.652855	-2.011345	1.10×10^{-4}	4.47×10^{-8}
64	0.60	ho	1.443005	-0.000047	1.443052	6.43×10^{-4}	1.55×10^{-8}

Table 5: (c) All 17 common half-line points, freshly computed with de Hoog and Clenshaw–Curtis.

Y	r	Fresh Toda anti	Continuum minus fresh	Quad. change	Saved difference	Nodes
4	0.40	0.760176674	-1.539453788	4.45×10^{-25}	5.08×10^{-28}	48
4	0.50	1.236044159	-1.761098636	3.15×10^{-20}	3.35×10^{-28}	48
4	0.60	1.474666481	-1.833155815	2.52×10^{-17}	2.46×10^{-28}	48
8	0.40	1.045500707	-1.824777821	3.30×10^{-18}	1.54×10^{-28}	48
8	0.50	1.402227994	-1.927282471	8.62×10^{-15}	2.54×10^{-25}	48
8	0.60	1.576516765	-1.935006100	4.49×10^{-13}	5.93×10^{-21}	48
16	0.40	1.178451129	-1.957728243	1.14×10^{-13}	9.98×10^{-23}	48
16	0.50	1.469719535	-1.994774011	4.17×10^{-11}	1.89×10^{-18}	48
16	0.60	1.621161074	-1.979650409	1.34×10^{-9}	4.57×10^{-16}	48
32	0.25	0.234074062	-1.821998909	3.87×10^{-15}	1.06×10^{-25}	48
32	0.40	1.234182737	-2.013459851	1.10×10^{-10}	1.56×10^{-16}	48
32	0.50	1.500789132	-2.025843609	8.20×10^{-9}	1.26×10^{-14}	48
32	0.60	1.642443888	-2.000933223	7.13×10^{-8}	9.62×10^{-13}	48
32	0.75	1.720935993	-1.909990649	1.91×10^{-10}	1.75×10^{-16}	96
64	0.40	1.260419907	-2.039697021	1.40×10^{-8}	1.04×10^{-12}	48
64	0.50	1.515812100	-2.040866577	8.66×10^{-11}	1.52×10^{-17}	96
64	0.60	1.652858825	-2.011348159	1.59×10^{-9}	1.14×10^{-15}	96

Table 6: (c) Independent free-intercept large-Y fits on the last 5, 6 and 4 saved scales respectively.

r	Quadratic / 5	Cubic / 6	Linear / 4	Range
0.4	1.2857717780	1.2857725529	1.2857936321	2.19×10^{-5}
0.5	1.5305276696	1.5305278252	1.5305354709	7.80×10^{-6}
0.6	1.6631322202	1.6631322668	1.6631358590	3.64×10^{-6}

Table 7: (c) New finite-h extrapolation choices at $Y = 64$; all 30 refits are retained in JSON.

r	Ch.	Package centre	Audit quad5	Audit cubic7	Audit change	Package h error
0.40	ov	1.260418654	1.260451023	1.260398839	5.22×10^{-5}	4.27×10^{-5}
0.40	ho	-0.000013983	-0.000172863	-0.000129988	4.29×10^{-5}	2.51×10^{-4}
0.50	ov	1.515809862	1.515717703	1.515776964	5.93×10^{-5}	7.09×10^{-5}
0.50	ho	-0.000026355	-0.000381922	-0.000215025	1.67×10^{-4}	4.09×10^{-4}
0.60	ov	1.652855357	1.652684921	1.652804441	1.20×10^{-4}	1.10×10^{-4}
0.60	ho	-0.000046788	-0.000578010	-0.000343261	2.35×10^{-4}	6.43×10^{-4}

(c) The fresh half-line run used 35-digit arithmetic; degree-32 versus degree-48 inverse transforms at three control points changed by at most 2.03×10^{-27}. The maximum final quadrature-refinement difference was 7.13×10^{-8}, while the maximum discrepancy from the saved half-line antisymmetric phase was 1.04×10^{-12}. These are sensitivity diagnostics for this model, not a rigorous global bound. The complete fresh audit run took 270.2792 seconds. Printed-table checks found 39 finite-amplitude and 48 ratio-table cells outside rounding error.

(c) Errors are kept separate: finite-sea h extrapolation, finite-sea mesh, half-line quadrature/inversion, and large-Y fit spread. The independent refits have free intercepts. The continuum formulas do not acquire a physical error bar merely by being evaluated to many digits. In particular neither the uncomputed determinant coefficient nor a missing quantum frame is an $O(\varepsilon^2)$ numerical uncertainty. All ratio-dependent discrepancies in (9) survive $Y \to \infty$; their size is far larger than the tested numerical sensitivity.

(d) V1 is therefore **not validated**. The rectangle leading coefficient passes a conditional consistency test. The two supplied canonical and directional phase representatives fail direct equality, even after a constant offset. Physical continuum–Toda equivalence remains inconclusive because these are not independently shown to be the same observable. This conclusion does not refute the good-equation programme or choose its renormalisation.

7 Coverage, numerical tables and failures recorded (13–14)

Script execution (c). The original three scripts were executed unchanged except that their single `Path.write_text` JSON output was intercepted in memory and restricted to the expected output path. No mathematical code, tolerance or input was altered. The measured wall times were respectively 39.1662, 5.9282, and 1.1479 seconds for `twobody_cont_phase.py`, `twobody_cont_det_antisym.py`, and `twobody_cont_grid.py`. Each exited zero and printed `ALL CHECKS PASSED`. Their success does not verify missing determinant integrals, projector assertions, printed tables, or the imported physical dictionary. No old phase-script bulk run was made.

(c) The saved grid's tested classical and canonical components are reproduced by independent computation to their serialized precision. The printed finite-amplitude table does not reproduce them: at $(0.1, 0.05)$ its overtaking classical phase is -0.016501, whereas independent computation and JSON give -0.0165462571730301; its H is 0.006635 versus 0.00668860438656757. At $r = 0.9$ the correct canonical limiting sum is -0.0673642536744, not the printed -0.074166. The checker enumerates all table discrepancies exceeding half a last printed digit. Most JSON fields are converted to Python floats, and selected string fields retain 35 significant digits, not 55-digit output accuracy. At $r = 0.1$ the serialized I already differs from independent quadrature by about 10^{-17}, disproving a blanket 10^{-50} output-uncertainty claim. Finite-amplitude determinant components are not tabulated at all. The sentence $D_{\mathrm{ov}}(1/2) = -3/(2\pi) \simeq -0.954930$ also has a factor-two error: the displayed decimal is $-3/\pi$.

Failures recorded.

1. **(c)** Refuted package checks/claims: negative bad profile and L_0; reversed stable/unstable bands; Toda sound speed; printed $A_{\mathrm{ov}}\ \varepsilon^4$ coefficient; bare canonical β^5 coefficient; incomplete general tadpole series; claimed finite-amplitude determinant verification; subleading rectangle; numerical tables and blanket output precision; and completion of V1. These discrepancies are retained as positive refutation assertions or explicit coverage findings, not hidden by passing the original scripts.

2. **(c)** A first audit run failed the unchanged absolute 10^{-55} action-gradient check at $\varepsilon = 10^{-6}$ because cancellation in the unfactored Hirota polynomial exhausted 70 digits. That check now uses 110 digits; the tolerance and unfactored method are unchanged.

3. **(c)** Two intermediate audit algebra targets failed: differentiating the specified lattice charges gives $ab\sinh\kappa/\kappa$, not the initially typed $ab\cosh\kappa$; the rectangle's coefficient of B/Y is $102 - 85\sqrt{2}$, not the initially typed $144 - 111\sqrt{2}$. Both were corrected from the independent symbolic derivatives/four-corner sum, with exact-zero assertions retained.

4. An exploratory inverse-Laplace call requested the Cohen algorithm, which this installed mpmath version does not implement. It raised `ValueError`; de Hoog inversion is used throughout the audit instead. No failed numerical result was accepted.

5. **(c)** The first independent edge run failed its unchanged $2 \cdot 10^{-7}$ quadrature-refinement threshold at $(Y, r) = (64, 0.5)$: the 24/48-node difference was $5.0990 \cdot 10^{-7}$. The rule is now refined to 96 nodes when needed, with the original threshold and the separate $2 \cdot 10^{-8}$ comparison check unchanged. Both the initial and final refinement differences are recorded.

6. The PDF skill's artifact-marker helper is absent from this workspace (`MODULE_NOT_FOUND`). The requested local LaTeX/PDF workflow proceeds with the installed compiler and visual inspection; no existing artifact is modified. Two late documentation patches initially missed their target context; neither changed a file, and the intended paragraph was subsequently inserted correctly. A search containing unescaped regex braces also failed and was rerun as a literal search; it had no numerical effect. The first build with generated tables failed because Python interpreted LaTeX backslashes as control characters in two headings. The headings were repaired without changing any numerical cell. An overbroad post-build log guard then matched the installed package's harmless description "Rerun checks for auxiliary files", preventing the PDF copy and therefore the next render. The guard now checks actual rerun-warning strings; both LaTeX compilations themselves had succeeded.

7. **(c)** The package's own recorded development failures were read. Its passing checks do not establish that the printed subsequent fixes were all propagated: the bare momentum, higher Hirota coefficient, determinant coverage and tables provide explicit counterexamples. No original tolerance was loosened by this audit.

Reproduction and limitations. Run:

```
python3 phase2/scripts/audit_twobody_cont.py
```

The final line is `ALL CHECKS PASSED`. The independent JSON records input hashes, individual edge runtimes, tables, fits and full-run time. The final coverage rerun took 53.0790 seconds with `--reuse-edge`, retaining the freshly computed 270.2792-second run's edge results after checking the same input hashes; no package cache supplies the independent edge calculation. Compile this new source with `pdflatex` twice. The final PDF was rendered and inspected. **(d)** A physical bad-equation contour/domain and the good equation's scheme remain PI-level research choices, outside the authorized audit; no preference-dependent choice has been made here.